Video Streaming Service Database Report

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1. Introduction

This project implements a video streaming service database using PostgreSQL and Microsoft Access. The system is designed to store and manage information related to actors, directors, videos, accounts, viewers, episodes, and watch history.

The database supports several important operations required by the project. These include retrieving the actors in a movie together with their pay and role, listing all episodes of a show, showing the watch history of viewers in an account for a given month, finding the impact of an actor on different genres of movies, and identifying the top videos watched in a month.

In addition to the backend database design, we also built a user-friendly Microsoft Access interface. This includes forms for data entry, reports for viewing results in a better format, and a start-up menu form that allows the user to navigate the application easily.

2. Project Directory Listing

The project directory listing was produced immediately before submission, as required in the project instructions. It includes all files used in the project, together with the date and time at which they were last saved.

Insert the PDF or screenshot of the full directory listing here.

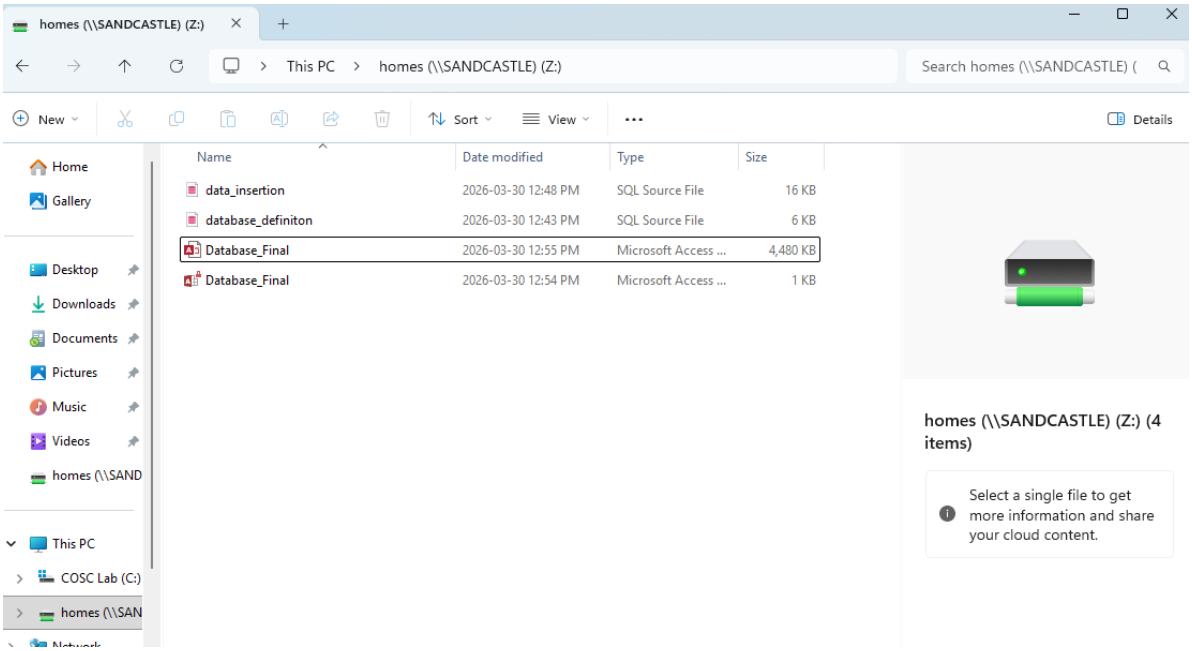

Figure 1: Full project directory listing

3. Database Schema

3.1 ER Model

The database was first designed using an Entity-Relationship model. The main entities in the system are Actor, Director, Video, Account, Viewer, and Episode. The main relationships include ActsIn, Directs, Watches, Shares, and BelongsTo. According to our Part 1 submission, the Video entity is specialized into Movie and Show. These subclasses are disjoint and not covering, meaning that a video can be either a movie or a show, but not both, and not every video must belong to one of these subclasses.

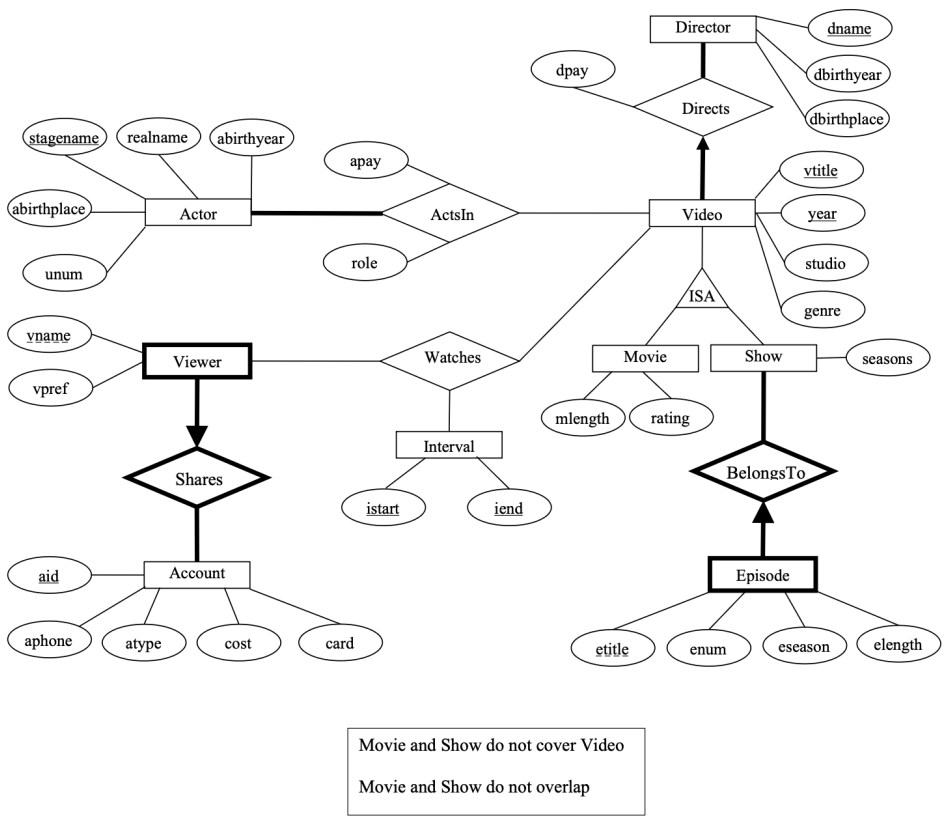

Figure 2: ER Diagram of the Video Streaming Service Database

3.2 Relational Schema

The ER model was converted into a relational schema. The final schema contains the following relations:

- Actor(stagename, realname, abirthyear, abirthplace, unum)
- Director(dname, dbirthyear, dbirthplace)
- AccountType(atype, mcost)
- Account(aid, aphone, ccard, atype)
- ViewerShares(vname, aid, vpref)
- VideoD(vtitle, year, studio, genre, dname, dpay)
- Movie(vtitle, year, mlength, rating)
- Show(vtitle, year, seasons)
- EpisodeOf(etitle, enum, eseason, vtitle, year, elength)
- ActsIn(stagename, vtitle, year, apay, role)
- Watches(vname, aid, vtitle, year, istart, iend)

The schema above follows the design created in Part 1. In that work, VideoD was formed by merging the Video entity with the Directs relationship, EpisodeOf was formed by merging Episode with BelongsTo, and ViewerShares was formed by merging Viewer with Shares.

3.3 Normalization

A functional dependency analysis was performed for all relations in Part 1. Based on that analysis, all relations satisfy BCNF because, in each case, the determinant of every functional dependency is a candidate key. Therefore, no further decomposition was required.

4. SQL Implementation and Constraints

The database was implemented in PostgreSQL using `CREATE TABLE` statements with primary keys, unique constraints, foreign keys, and check constraints. In addition, triggers were used to enforce some advanced rules such as valid episode season ranges, viewer limits for standard accounts, and prevention of overlapping watch intervals.

Below is the file path for SQL implementation used for the database schema.

File Path:

`"Z:\database_definiton.sql"`

5. Table Data

This section shows the data stored in each table. The project instructions require a list of all data for each table. We include screenshots of each table after inserting the sample data.

5.1 Actor

public_actor

stagename	realname	abirthyear	abirthplace	unum
Matt Patel	Matthew Patel	1970	Cambridge MA	1002
Leo DiCaprio	Leonardo DiCaprio	1974	Los Angeles CA	1001
Margot Robbie	Margot Elise Robbie	1990	Dalby QLD	1003
Ryan Gosling	Ryan Thomas Gosling	1980	London Ontari	1004
Jack Nicholson	John Joseph Nicholson	1937	Neptune City N	1006
Bryan Cranston	Bryan Lee Cranston	1956	Hollywood CA	1005
Timothee Chalamet	Timothee Hal Chalamet	1995	New York NY	2001
Zendaya	Zendaya Coleman	1996	Oakland CA	2002
Brad Pitt	William Bradley Pitt	1963	Shawnee OK	2003
Samuel L Jackson	Samuel Leroy Jackson	1948	Washington DC	2004
Kate Winslet	Kate Elizabeth Winslet	1975	Reading UK	2005
Ellen Pompeo	Ellen Kathleen Pompeo	1969	Everett MA	2006
Millie Bob	Millie Bobby Brown	2004	Marbella Spair	1007

Figure 3: Contents of Actor table

5.2 Director

public_director

dname	dbirthyear	dbirthplace
Christopher N	1970	London Englan
Greta Gerwig	1983	Sacramento CA
Martin Scorses	1942	New York NY
David Lynch	1946	Missoula MT
Vince Gilligan	1967	Farmville VA
Matt Duffer	1984	Durham NC
Denis Villeneu	1967	Gentilly Canad
Quentin Taran	1963	Knoxville TN
James Camero	1954	Kapuskasing O
Shonda Rhime	1970	Chicago IL

Figure 4: Contents of Director table

5.3 AccountType

public_account | public_accounttype

atype	cost
Standard	9.99
Premium	15.99

Figure 5: Contents of AccountType table

5.4 Account

public_account

aid	aphone	atype	card
4	4161110004	Standard	4100000000000
5	4161110005	Premium	4100000000000
6	4161110006	Standard	4100000000000
7	4161110007	Standard	4100000000000
8	4161110008	Standard	4100000000000
1	4161110001	Standard	4100000000000
2	4161110003	Premium	4100000000000
3	123456789	Standard	1111
9	4162220001	Standard	4200000000000
10	4162220002	Premium	4200000000000
11	4162220003	Standard	4200000000000
12	4162220004	Premium	4200000000000

Figure 6: Contents of Account table

5.5 ViewerShares

public_viewershares

vname	vpref	aid
Alice	Action	1
Eve	Mystery	4
Frank	Comedy	5
Grace	Drama	5
Henry	Sci-Fi	6
Ivy	Romance	7
Jack	Action	8
Carol	Comedy	2
Dave	Crime	2
Bob	Sci-Fi	3
Liam	Sci-Fi	9
Noah	Drama	10
Olivia	Romance	10
Emma	Action	11
Ava	Comedy	12
Sophia	Drama	12
Vraj	Crime	5

Figure 7: Contents of ViewerShares table

5.6 VideoD

public_show | public_videod

vtitle	year	studio	genre	dname	dpay
Inception	2010	Warner Bros	Sci-Fi	Christopher N	20000000
The Departed	2006	Warner Bros	Crime	Martin Scorses	18000000
Mulholland Dr	2001	Universal	Mystery	David Lynch	9000000
Goodfellas	1990	Warner Bros	Crime	Martin Scorses	15000000
Breaking Bad	2008	AMC Studios	Crime	Vince Gilligan	5000000
Stranger Thing	2016	Netflix	Sci-Fi	Matt Duffer	8000000
Dune	2021	Warner Bros	Sci-Fi	Denis Villeneu	18000000
Pulp Fiction	1994	Miramax	Crime	Quentin Taran	15000000
Titanic	1997	Paramount	Romance	James Camero	20000000
Avatar	2009	Fox	Sci-Fi	James Camero	22000000
Grey Anatomy	2005	ABC Studios	Drama	Shonda Rhime	5000000
Interstellar	2014	Paramount	Sci-Fi	Christopher N	20000000
Barbie	2023	Warner Bros	Comedy	Greta Gerwig	15000000

Figure 8: Contents of VideoD table

5.7 Movie

public_movie

vtitle	year	mlength	rating
Inception	2010	148	87
Barbie	2023	114	75
The Departed	2006	151	90
Mulholland Dr	2001	147	82
Goodfellas	1990	145	95
Interstellar	2014	169	86
Dune	2021	155	88
Pulp Fiction	1994	154	94
Titanic	1997	195	89
Avatar	2009	162	85

Figure 9: Contents of Movie table

5.8 Show

public_show

vtitle	year	seasons
Breaking Bad	2008	2
Stranger Thing	2016	1
Grey Anatomy	2005	2

Figure 10: Contents of Show table

5.9 EpisodeOf

public_episodeof

etitle	enum	eseason	elength	vtitle	year
Pilot	1	1	58	Breaking Bad	2008
Cats in the Bag	2	1	48	Breaking Bad	2008
Bag in the Rive	3	1	48	Breaking Bad	2008
Seven Thirty-S	1	2	47	Breaking Bad	2008
Down	2	2	47	Breaking Bad	2008
Will Byers Mis:	1	1	47	Stranger Thing	2016
Weirdo on Ma	2	1	55	Stranger Thing	2016
Holly Jolly	3	1	50	Stranger Thing	2016
A Hard Day	1	1	42	Grey Anatomy	2005
First Cut	2	1	43	Grey Anatomy	2005
Into You	1	2	44	Grey Anatomy	2005

Figure 11: Contents of EpisodeOf table

5.10 ActsIn

public_actsin

stagename	vtitle	year	apay	role
Leo DiCaprio	Inception	2010	20000000	Dom Cobb
Leo DiCaprio	The Departed	2006	20000000	Billy Costigan
Leo DiCaprio	Goodfellas	1990	1000000	Henry Hill
Jack Nicholson	The Departed	2006	10000000	Frank Costello
Jack Nicholson	Mulholland Dr	2001	5000000	Adam Kesher
Margot Robbie	Barbie	2023	12500000	Barbie
Ryan Gosling	Barbie	2023	12000000	Ken
Bryan Cranstor	Breaking Bad	2008	225000	Walter White
Matt Patel	The Departed	2006	15000000	Colin Sullivan
Matt Patel	Interstellar	2014	15000000	Dr. Mann
Jack Nicholson	Goodfellas	1990	8000000	Jimmy Conway
Matt Patel	Breaking Bad	2008	200000	Guest Star
Timothee Chal	Dune	2021	10000000	Paul Atreides
Zendaya	Dune	2021	8000000	Chani
Brad Pitt	Pulp Fiction	1994	12000000	Vincent
Samuel L Jacks	Pulp Fiction	1994	11000000	Jules
Kate Winslet	Titanic	1997	9000000	Rose
Kate Winslet	Avatar	2009	10000000	Ronald
Millie Bob	Stranger Thing	2016	300000	Eleven

Figure 12: Contents of ActsIn table

5.11 Watches

public_viewershares | public_watches

vname	aid	vtitle	year	istart	iend
Alice	1	Inception	2010	2026-01-05 10:00:00 AM	2026-01-05 12:00:00 PM
Alice	1	Breaking Bad	2008	2026-01-10 8:00:00 PM	2026-01-10 9:00:00 PM
Alice	1	Inception	2010	2026-01-15 8:00:00 PM	2026-01-15 10:00:00 PM
Eve	4	Inception	2010	2026-01-12 2:00:00 PM	2026-01-12 4:00:00 PM
Eve	4	The Departed	2006	2026-01-25 8:00:00 PM	2026-01-25 10:30:00 PM
Frank	5	Barbie	2023	2026-01-06 6:00:00 PM	2026-01-06 8:00:00 PM
Frank	5	Interstellar	2014	2026-01-18 2:00:00 PM	2026-01-18 4:30:00 PM
Grace	5	Barbie	2023	2026-01-07 2:00:00 PM	2026-01-07 4:00:00 PM
Grace	5	Interstellar	2014	2026-01-20 4:00:00 PM	2026-01-20 6:30:00 PM
Henry	6	Barbie	2023	2026-01-10 4:00:00 PM	2026-01-10 6:00:00 PM
Henry	6	Interstellar	2014	2026-01-22 2:00:00 PM	2026-01-22 4:30:00 PM
Ivy	7	Barbie	2023	2026-01-14 8:00:00 PM	2026-01-14 10:00:00 PM
Ivy	7	Interstellar	2014	2026-01-28 2:00:00 PM	2026-01-28 4:30:00 PM
Jack	8	Interstellar	2014	2026-01-30 2:00:00 PM	2026-01-30 4:30:00 PM
Alice	1	Barbie	2023	2026-02-14 7:00:00 PM	2026-02-14 9:00:00 PM
Eve	4	Interstellar	2014	2026-02-08 3:00:00 PM	2026-02-08 5:30:00 PM
Frank	5	Inception	2010	2026-02-12 10:00:00 AM	2026-02-12 12:00:00 PM
Grace	5	Inception	2010	2026-02-15 11:00:00 AM	2026-02-15 1:00:00 PM
Henry	6	The Departed	2006	2026-02-18 7:00:00 PM	2026-02-18 9:30:00 PM
Ivy	7	Barbie	2023	2026-02-22 7:00:00 PM	2026-02-22 9:00:00 PM
Jack	8	The Departed	2006	2026-02-25 8:00:00 PM	2026-02-25 10:30:00 PM
Alice	1	Interstellar	2014	2026-03-15 2:00:00 PM	2026-03-15 4:30:00 PM
Eve	4	Inception	2010	2026-03-05 10:00:00 AM	2026-03-05 12:00:00 PM
Frank	5	Breaking Bad	2008	2026-03-08 8:00:00 PM	2026-03-08 9:00:00 PM
Grace	5	Goodfellas	1990	2026-03-12 8:00:00 PM	2026-03-12 10:00:00 PM
Henry	6	Inception	2010	2026-03-15 10:00:00 AM	2026-03-15 12:00:00 PM
Ivy	7	Breaking Bad	2008	2026-03-18 8:00:00 PM	2026-03-18 9:00:00 PM
Jack	8	Inception	2010	2026-03-20 2:00:00 PM	2026-03-20 4:00:00 PM
Carol	2	Barbie	2023	2026-02-20 3:00:00 PM	2026-02-20 5:00:00 PM
Carol	2	Breaking Bad	2008	2026-01-15 9:00:00 PM	2026-01-15 10:00:00 PM
Carol	2	Inception	2010	2026-01-08 6:00:00 PM	2026-01-08 8:00:00 PM
Carol	2	Interstellar	2014	2026-03-10 1:00:00 PM	2026-03-10 3:30:00 PM
Dave	2	Barbie	2023	2026-03-20 6:00:00 PM	2026-03-20 8:00:00 PM
Dave	2	Goodfellas	1990	2026-02-05 8:00:00 PM	2026-02-05 10:00:00 PM
Dave	2	Inception	2010	2026-01-09 8:00:00 PM	2026-01-09 10:00:00 PM
Dave	2	The Departed	2006	2026-01-22 7:00:00 PM	2026-01-22 9:30:00 PM
Bob	3	Breaking Bad	2008	2026-02-10 7:00:00 PM	2026-02-10 8:00:00 PM

Figure 13: Contents of Watches table

6. Required Queries

This section contains the five required queries for the project together with a short explanation of what each query does.

6.1 Query 1: Actors in a Given Movie

This query returns the stage name, pay, and role of all actors in a given movie, using the movie title and year as input.

```
SELECT A.stagename, AI.apay, AI.role
FROM public_actor AS A, public_actsin AS AI, public_movie AS M
WHERE A.stagename = AI.stagename
  AND AI.vtitle = M.vtitle
  AND AI.[year] = M.[year]
  AND M.vtitle = "Inception"
  AND M.[year] = 2010;
```

6.2 Query 2: All Episodes of a Given Show

This query returns all episode information for a given show, based on the title and year of the show.

```
SELECT E.etitle, E.enum, E.eseason, E.elength, E.vtitle, E.[year]
FROM public_episodeof AS E, public_show AS S
WHERE E.vtitle = S.vtitle
  AND E.[year] = S.[year]
  AND S.vtitle = "Breaking Bad"
  AND S.[year] = 2008
ORDER BY E.eseason, E.enum;
```

6.3 Query 3: Watch History for an Account in a Given Month

This query returns all viewers belonging to a chosen account together with their watch history in a selected month.

```
SELECT VS.vname, W.istart, W.iend, W.vtitle, W.[year]
```

```
FROM public_viewershares AS VS, public_watches AS W
WHERE VS.vname = W.vname
  AND VS.aid = W.aid
  AND VS.aid = 3
  AND W.istart >= #01/01/2026 00:00:00#
  AND W.istart <  #02/01/2026 00:00:00#
ORDER BY VS.vname, W.istart;
```

6.4 Query 4: Actor Impact on Genres

This query finds the average rating of movies in each genre in which a chosen actor has acted.

```
SELECT V.genre, AVG(M.rating) AS avg_rating
FROM public_actsin AS AI, public_movie AS M, public_videod AS V
WHERE AI.vtitle = M.vtitle
  AND AI.[year] = M.[year]
  AND M.vtitle = V.vtitle
  AND M.[year] = V.[year]
  AND AI.stagename = "Leo DiCaprio"
GROUP BY V.genre
ORDER BY V.genre;
```

6.5 Query 5: Top 5 Videos in a Given Month

This query returns the top 5 videos in a given month, together with the number of different viewers who watched each video.

```
SELECT TOP 5 Q.vtitle, Q.[year], COUNT(*) AS total_viewers
FROM
(
    SELECT DISTINCT W.vtitle, W.[year], W.vname, W.aid
    FROM public_watches AS W
    WHERE W.istart >= #01/01/2026 00:00:00#
      AND W.istart <  #02/01/2026 00:00:00#
) AS Q
GROUP BY Q.vtitle, Q.[year]
ORDER BY COUNT(*) DESC, Q.vtitle, Q.[year];
```

7. Forms and User Interface

To make the system easier to use, Microsoft Access forms were created for the main entities and relationships. These forms allow users to enter, edit, and view data in a cleaner way than working directly with tables.

We also created a start-up menu form that opens when the application starts. This menu allows the user to open forms, queries, and reports from one place. This makes the project more user-friendly and also helps satisfy the bonus requirements related to interface and layout.

7.1 Main Menu Form

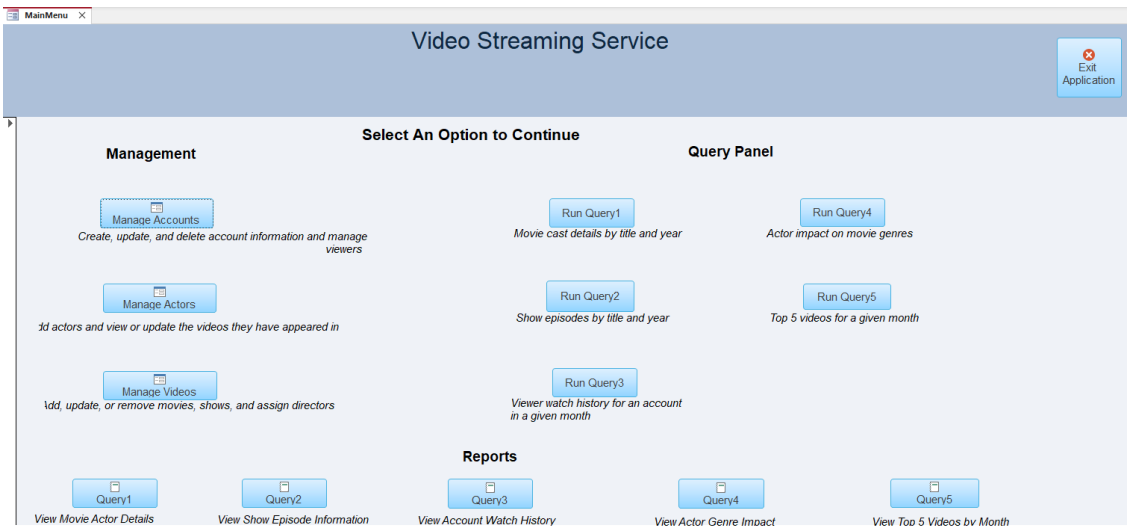

Figure 14: Main menu form

7.2 Example Data Entry Forms

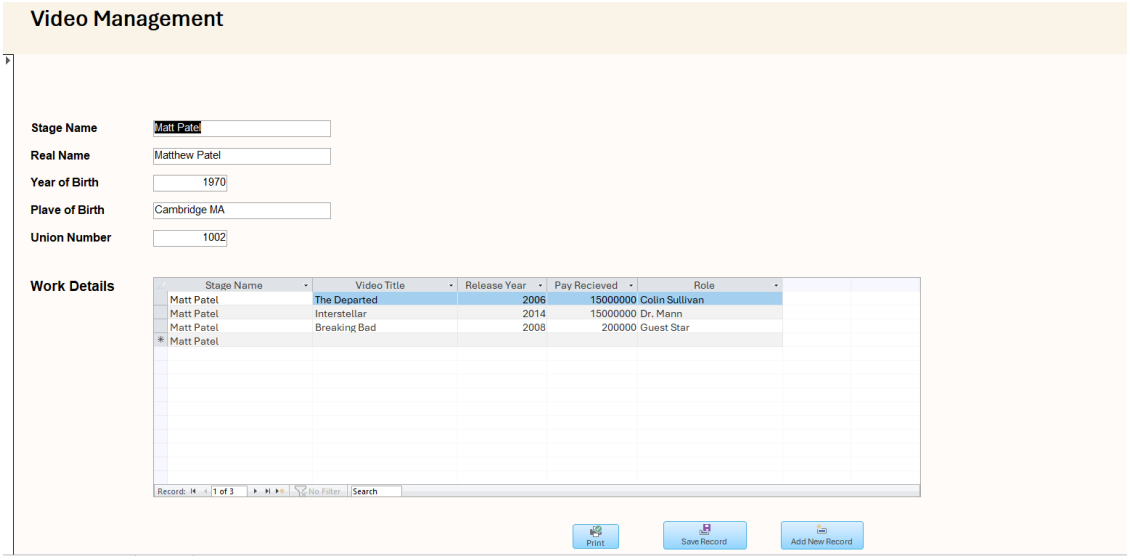

Figure 15: Actor form

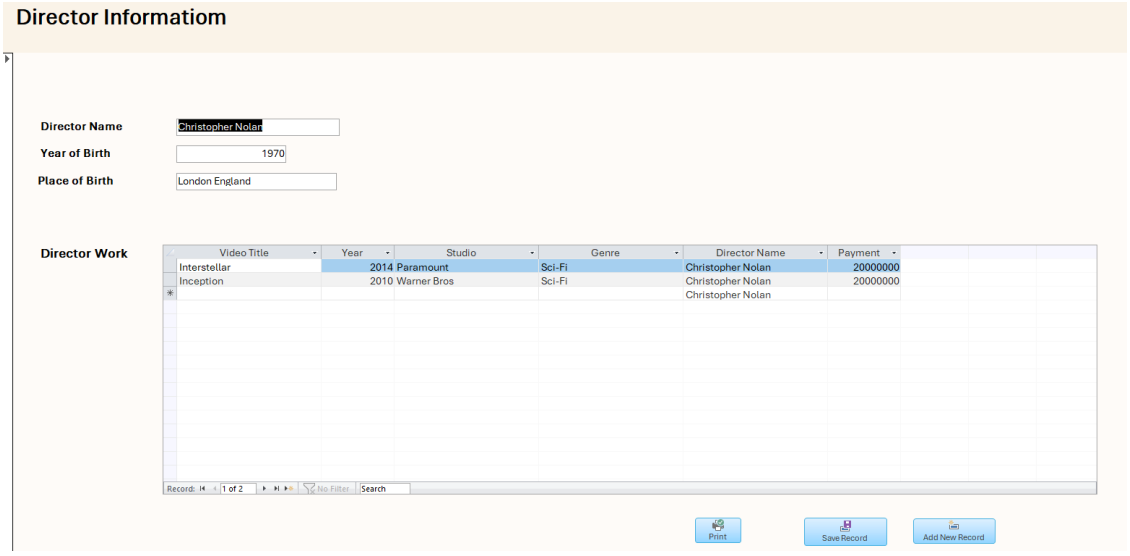

Figure 16: Movie form

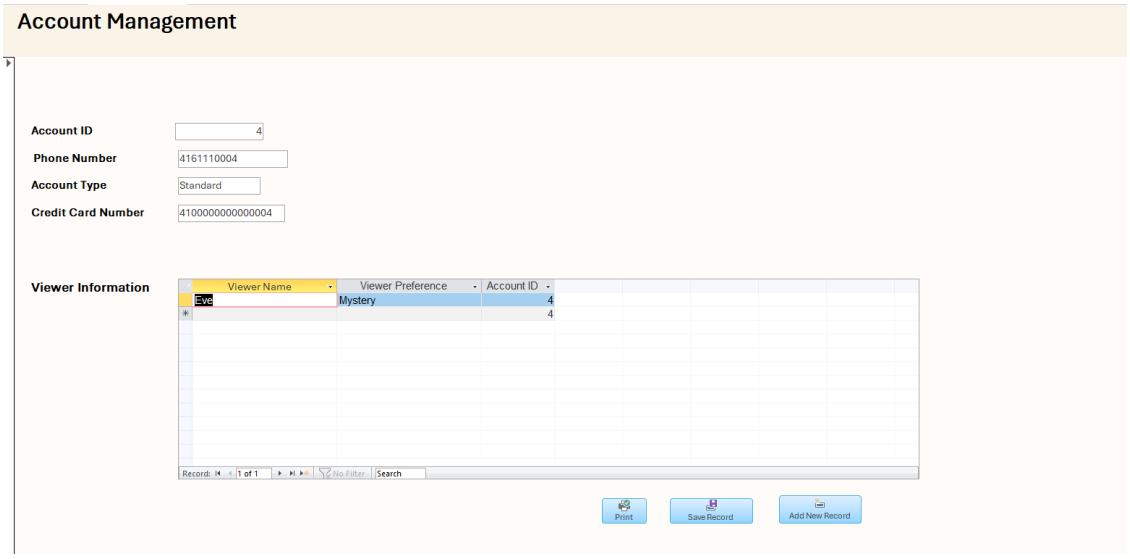

Figure 17: Watches form

8. Reports

To improve presentation and usability, Access reports were created for the required queries. These reports display query results in a cleaner and more readable form than raw query output.

8.1 Report 1: Actors in a Movie

Figure 18: Report for Query 1

8.2 Report 2: Episodes of a Show

Query2

etitle	enum	eseason	elength	vtitle	year
Bag in the River					
	3	1	48	Breaking Bad	2008
Cats in the Bag					
	2	1	48	Breaking Bad	2008
Down					
	2	2	47	Breaking Bad	2008
Pilot					
	1	1	58	Breaking Bad	2008
Seven Thirty-Seven					
	1	2	47	Breaking Bad	2008

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Figure 19: Report for Query 2

8.3 Report 3: Watch History

Query3

vname	vtitle	year	istart	iend
Carol				
	Breaking Bad			
		2008	2026-01-15 9:00:00 PM	2026-01-15 10:00:00 PM
	Inception			
		2010	2026-01-08 6:00:00 PM	2026-01-08 8:00:00 PM
Dave				
	Inception			
		2010	2026-01-09 8:00:00 PM	2026-01-09 10:00:00 PM
	The Departed			
		2006	2026-01-22 7:00:00 PM	2026-01-22 9:30:00 PM

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Figure 20: Report for Query 3

8.4 Report 4: Actor Genre Impact

Query4

genre	avg_rating
Crime	
	92.5
Sci-Fi	
	87

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Figure 21: Report for Query 4

8.5 Report 5: Top 5 Videos

Query5

vtitle	total_viewers	year
Barbie		
	4	2023
Breaking Bad		
	2	2008
Inception		
	5	2010
Interstellar		
	5	2014
The Departed		
	3	2006

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Figure 22: Report for Query 5

9. Bonus Features

In addition to the required project functionality, several extra features were implemented to improve the quality of the project.

- A start-up menu form was created so that the application opens to a clear main menu.
- Forms were designed with a cleaner layout to make data entry easier.In addition, buttons are provided with Print, Save/Submit, and Add New Record
- Reports were created for the required queries to provide better formatted output features.
- The interface was made more user-friendly by adding buttons for forms, queries, reports, and exiting the application.

These features make the project easier to use and help satisfy the bonus requirements in the project instructions.